

18-874: Spacecraft Design-Build-Fly Lab

Spring 2026

Course Description

Spacecraft design is a truly interdisciplinary subject that draws from every branch of engineering. This capstone design class brings together the material from prior classes in a way that emphasizes the interactions between disciplines and demonstrates how some of the more theoretical topics are synthesized in the practical design of a spacecraft. The class will design, build, and test a small satellite that addresses objectives and requirements posed at the beginning of the course sequence. Students will work in subsystem teams, each focusing on some aspect of the spacecraft, but exposed to many different disciplines and challenges. Practical, hands-on engineering skills will be emphasized, along with fabrication and testing of physical hardware and the creation of thorough documentation.

Overview for Fall 2026:

In this offering of the course the team will be developing and implementing features for Argus-3, based on the design of Argus-2, which is currently in the final process of launch logistics for a future launch in 2027/2028 timeframe. These Argus-3 features will be designed, developed and integrated into a future-looking variant of the Argus flight bus that will fly shortly after the Argus-2 bus.

Instructors

Prof. Brandon Lucia

System Scientist & Labmaster: Neil Khera

TA: Sun A Cho

TA: Om Anavekar

TA: Veronica Muriga

Email: blucia@andrew.cmu.edu

Email: nkhera@andrew.cmu.edu

Email: sunac@andrew.cmu.edu

Email: oanaveka@andrew.cmu.edu

Email: vmuriga@andrew.cmu.edu

Learning Objectives

The goal of this course is to give students hands-on experience designing and building small spacecraft subsystems and integrating them into a CubeSat. Throughout this course, students will:

1. Understand how the design and integration of a system whose performance depends on the success of many interacting subsystems.
2. Work within a small team to fabricate, and test hardware and software through rapid design iteration.
3. Coordinate with other teams to integrate subsystems into a complete spacecraft.
4. Gain exposure to the complete life cycle of a small satellite mission.

5. Develop skills at project management and systems engineering
6. Develop social skills and planning skills to work with a large group and coordinate many required tasks to launch-ready a satellite.

Logistics

The course will involve designing and building hardware in small teams. Class time will be used primarily for weekly team meetings and consulting time to meet with the instructors.

- Class sessions: MW 3:30-4:50
- All-hands meetings will be held every week on Wednesdays, followed by consulting hours.
- Mondays will be in-lab working sessions unless there is a lecture scheduled, in which case lecture will be in PH A18A
- Students are expected to be in lab during in-lab working sessions and outside of schedule course hours.
- Slack will be used for coordination between teams and instructors. All students will be added to the “SpacecraftDesignBuildFlyLab” slack channel.
- GitHub will be used to manage project files for all teams and teams will be required to use good git hygiene and software engineering practices.

Assignments and Exams

There will be no exams in this course. Evaluation will be based on participation, contribution to design and fabrication work, final documentation, and weekly project management metrics from each team. Grade details are broken down below. Use of project management tools is mandatory and your grade will be derived from specific documentation of technical contributions.

You will keep an individual weekly lab notebook. This will document your progress and activities for the week, every week. Your team will additionally keep an all-hands presentation log that gets updated on a weekly cadence as well.

Grading

Grading will be based on:

- 20% Participation and attendance of team meetings
- 30% Individual technical contributions quantified by git commit history / number and significance of PRs and peer surveys
- 30% Completeness and quality of team documentation
- 20% Final presentation and report on team progress for semester

Learning Resources

There is no textbook required for this course.

Course Policies

Attendance: This is a team-based course. In order to coordinate work among teams, participation in weekly meetings is required. If you are unable to be present at a meeting, you must notify the instructors and ensure that your teammates are prepared to present your work.

Accommodations for Students with Disabilities: If you have a disability and are registered with the Office of Disability Resources, I encourage you to use their online system to notify me of your accommodations and discuss your needs with me as early in the semester as possible. I will work with you to ensure that accommodations are provided as appropriate. If you suspect that you may have a disability and would benefit from accommodations but are not yet registered with the Office of Disability Resources, I encourage you to contact them at access@andrew.cmu.edu.

Statement of Support for Students' Health & Well-Being: Take care of yourself. Do your best to maintain a healthy lifestyle this semester by eating well, exercising, avoiding drugs and alcohol, getting enough sleep, and taking some time to relax. This will help you achieve your goals and cope with stress.

If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support. Counseling and Psychological Services (CaPS) is here to help: call 412-268-2922 and visit <http://www.cmu.edu/counseling>. Consider reaching out to a friend, faculty, or family member you trust for help getting connected to the support that can help.

If you or someone you know is feeling suicidal or in danger of self-harm, call someone immediately, day or night:

CaPS: 412-268-2922

Re:solve Crisis Network: 888-796-8226

If the situation is life threatening, call the police:

On campus: CMU Police: 412-268-2323

Off campus: 911

Tentative Fall 2026 Schedule

Week	Dates	Topics
1	Aug 24 Aug 26	Course Overview & Logistics Mission PDR Review and Team Assignments
2	Aug 31 Sep 2	Project Management All-hands Meeting
3	Sep 7 Sep 9	No Class (Labor Day) All-Hands Meeting
4	Sep 14 Sep 16	Lecture Orbital Edge Computing All-Hands Meeting
5	Sep 21 Sep 23	No Class (Instructor Travel) All-Hands Meeting
6	Sep 28 Sep 30	Lecture Efficient's Dataflow Architecture All-Hands Meeting
7	Oct 5 Oct 7	Case Study: Tartan Artibeus All-Hands Meeting
8	Oct 12 Oct 14	No Class (Spring Break) No Class (Spring Break)
9	Oct 19 Oct 21	Guest Lecture: Paulo Fisch All-Hands Meeting
10	Oct 26 Oct 28	Case Study: PROMETHEUS-1 (Neil Khera) All-Hands Meeting
11	Nov 2 Nov 4	Lab Work Day All-Hands Meeting
12	Nov 9 Nov 11	Guest Lecture or Lab Day All-Hands Meeting
13	Nov 16 Nov 18	No Class (Instructor Travel) All-Hands Meeting (Prof. Lucia out)
14	Nov 23 Nov 25	Final presentation feedback and final lab day No Class (Thanksgiving week)
14	Nov 30 Dec 2	Final Presentations / Design Review Final Presentations / Design Review